

Decode 2026

Reproducible Data Science with Git and GitHub

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July 20 2026

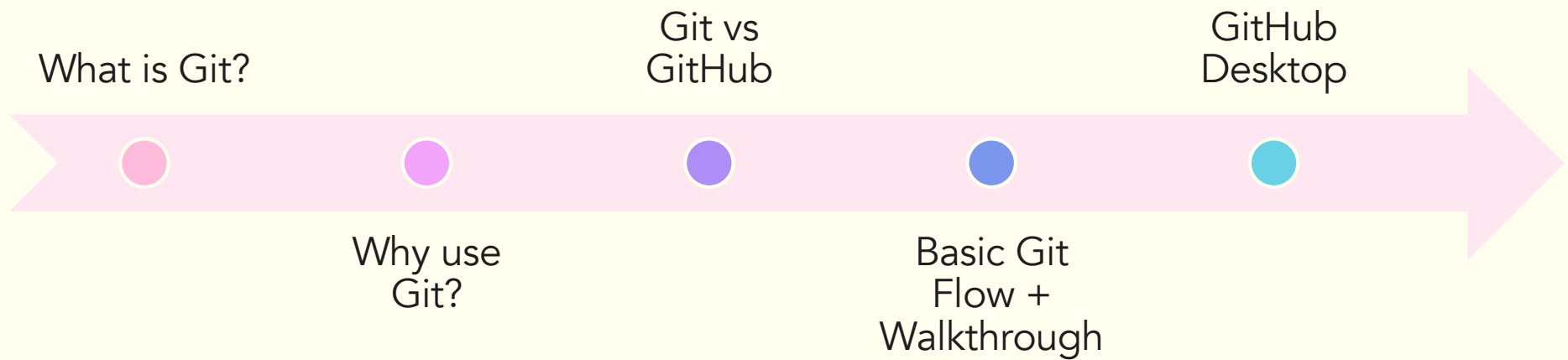
Where are you zooming in from?

2

Drop your city/state in the chat!

Today's focus

3



Disaster strikes when:

4

Your computer
crashes

You change
code and things
break later on

You want to
experiment with
your code

You need to
collaborate on a
group project

Version control to the rescue!

5

Think Google Docs but for code!

Every change made to the code base is tracked

Saves snapshots of your code

Lets you easily share code with the team you are working with

Allows you to collaborate on data projects and know who contributed to the changes made

Gives you access to any previous version of your document

***Git** is a commonly used version control system*

Quick example

6

In our last workshop, we explored data manipulation with R using the MoTrPAC data

Key Takeaways

The Six Essential dplyr Functions:

1. `select()`: Choose which columns to keep or remove
2. `filter()`: Choose which rows to keep based on conditions
3. `arrange()`: Sort your data by one or more columns
4. `mutate()`: Create new variables or modify existing ones
5. `summarise()`: Calculate summary statistics
6. `group_by()`: Group data for group-specific operations

Quick example

7

In our last workshop, we explored data manipulation with R using the MoTrPAC data

Key Takeaways

The Six Essential dplyr Functions:

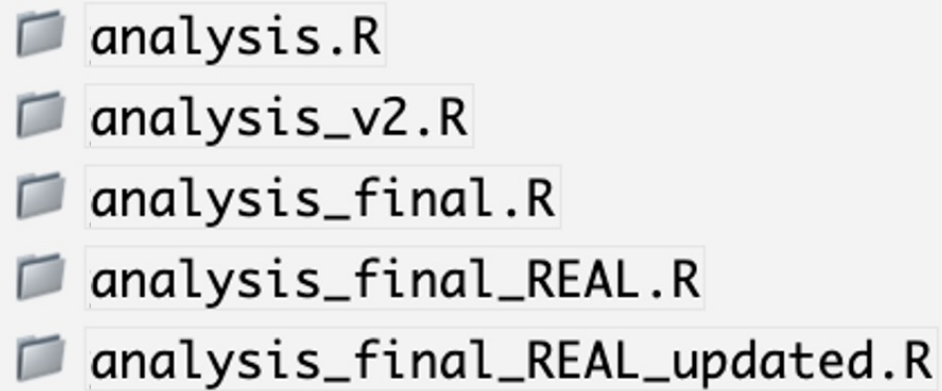
1. `select()`: Choose which columns to keep or remove
2. `filter()`: Choose which rows to keep based on conditions
3. `arrange()`: Sort your data by one or more columns
4. `mutate()`: Create new variables or modify existing ones
5. `summarise()`: Calculate summary statistics
6. `group_by()`: Group data for group-specific operations

What if your script breaks here and you want to go back to step 2 of your analysis?

File Name Chaos

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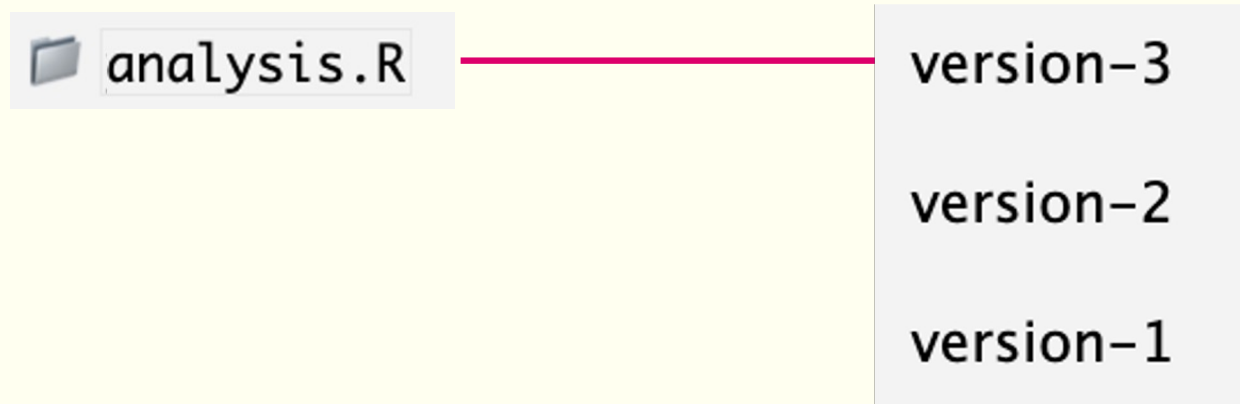
What if you made additional analyses and updates to your script?



- analysis.R
- analysis_v2.R
- analysis_final.R
- analysis_final_REAL.R
- analysis_final_REAL_updated.R

With Git

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Every version is
saved
automatically!

<https://biodatasage.com/fair>

Git VS GitHub

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Git (local)

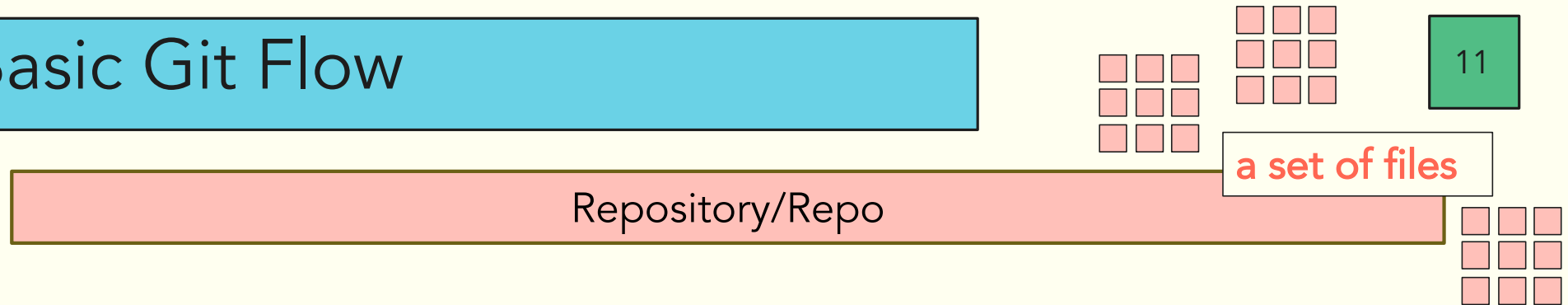
- Lives on your computer
- Tracks your changes
- Works offline
- Personal backup

GitHub (online)

- Lives on the internet
- Stores your Git repositories
- Share with others
- Online backup

<https://github.com>

Basic Git Flow



A folder for your programming project that can contain:

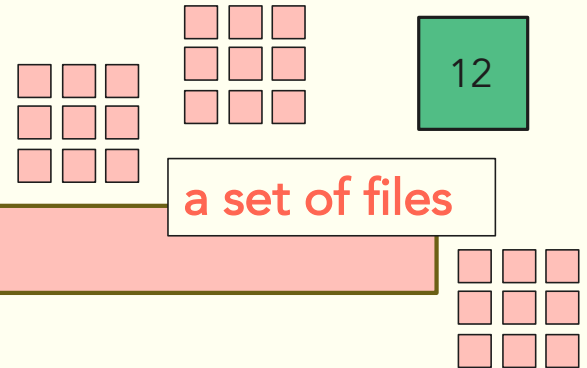
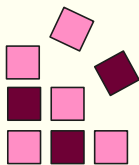
1. R scripts (or any other programming language)
2. Any datasets used (e.g. CSV files, excel files)
3. Any outputs generated during your analysis
4. Any other documents related to your project

Basic Git Flow

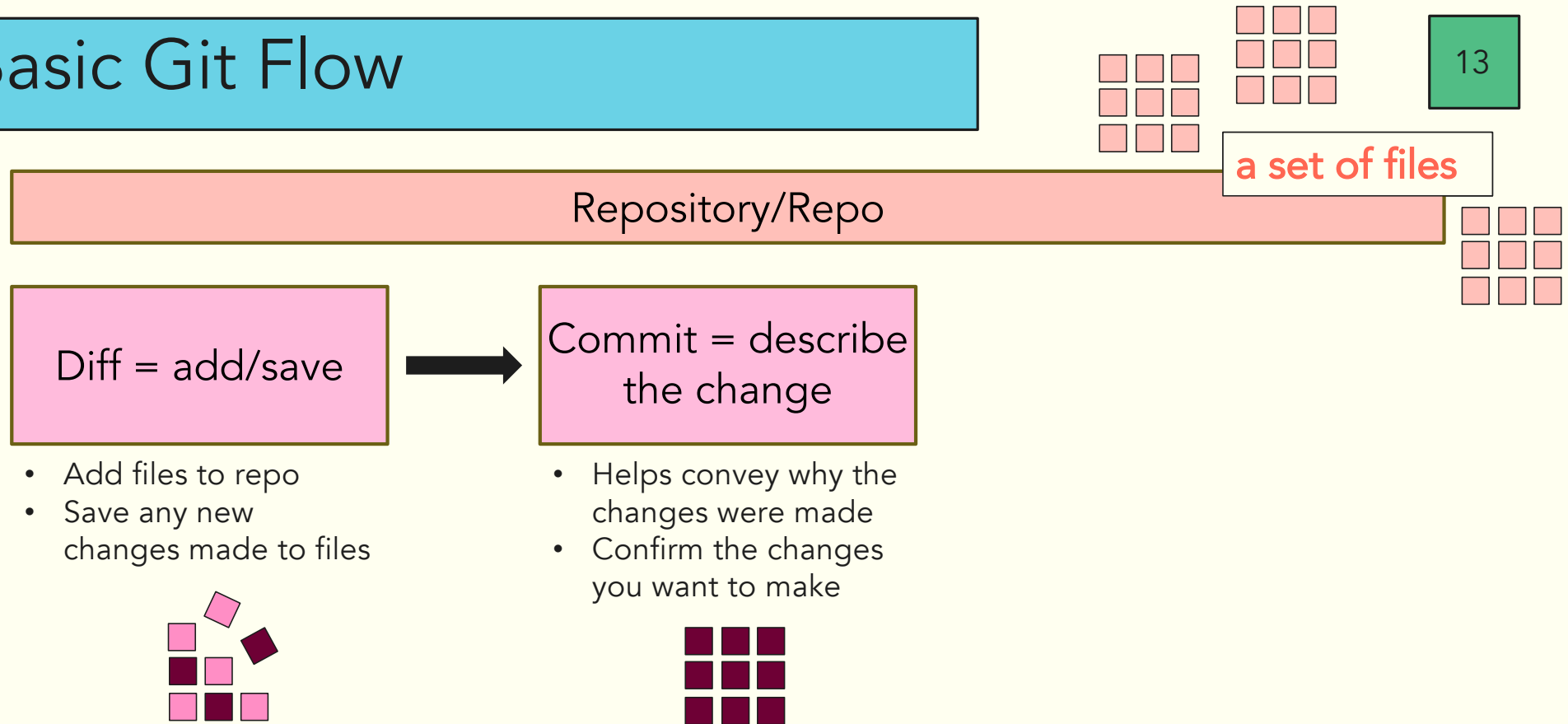
Repository/Repo

Diff = add/save

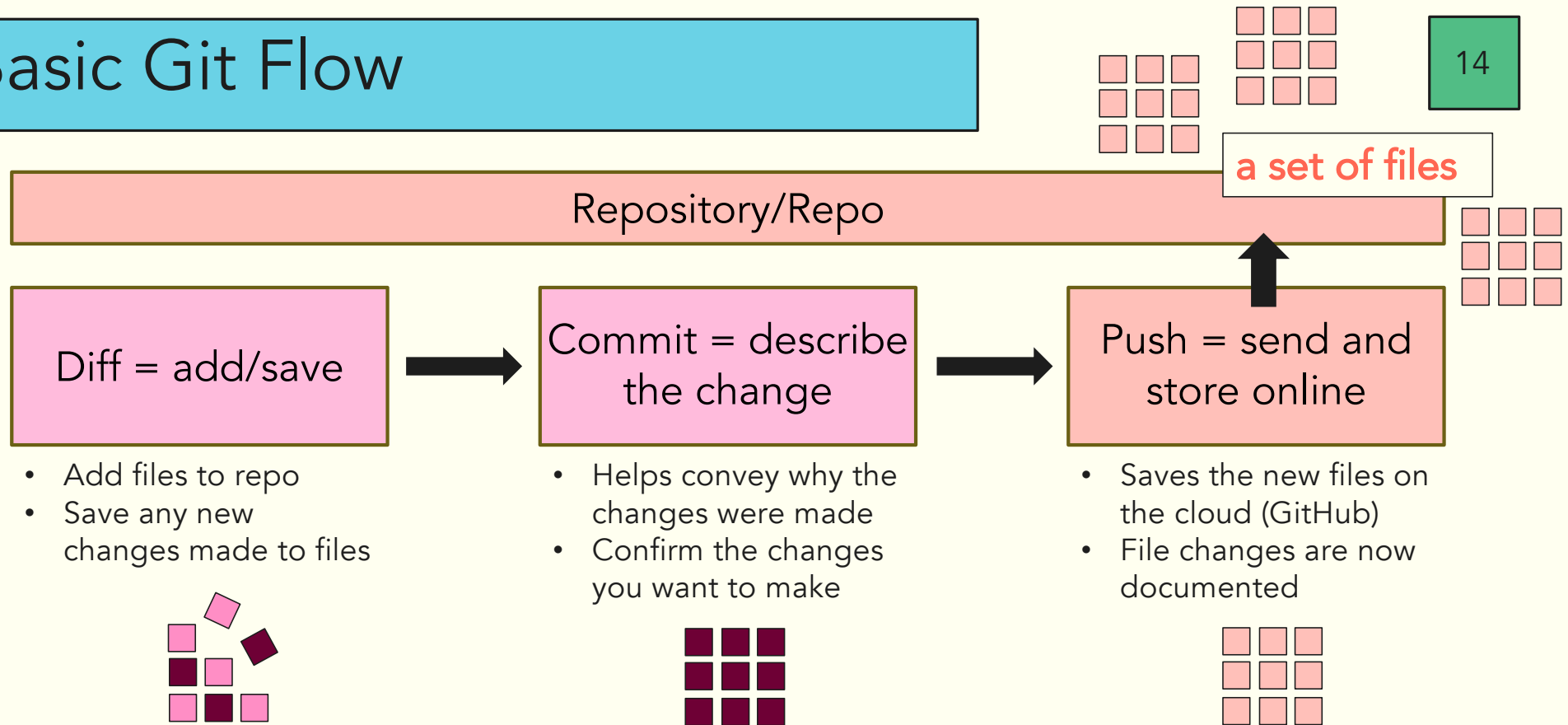
- Add files to repo
- Save any new changes made to files



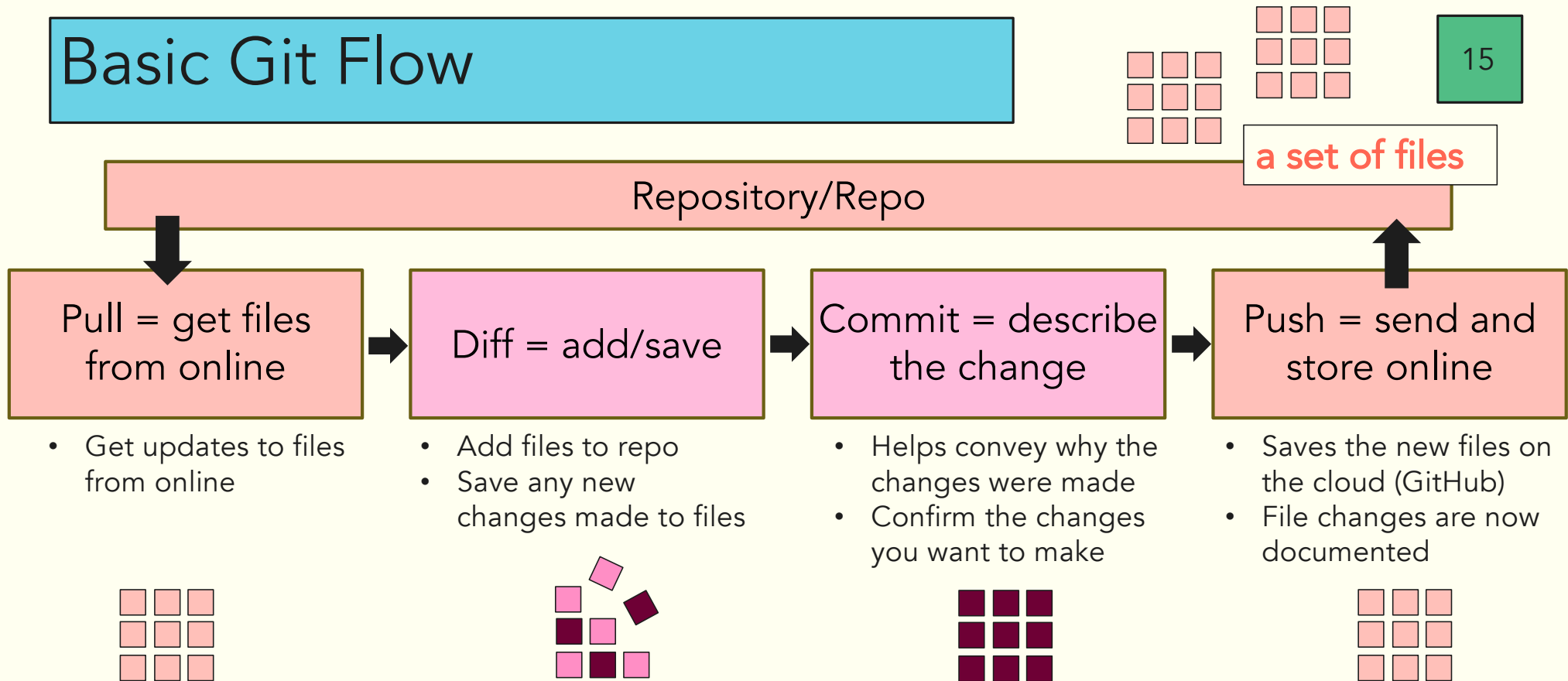
Basic Git Flow



Basic Git Flow



Basic Git Flow



How do we do all of this
on RStudio?

The screenshot shows the RStudio IDE interface with several annotations:

- Write notes here: A pink box highlights the top right area of the editor, containing a list of topics:

 - Differential analysis results, e.g., HEART_PROT_DA
 - Sample outliers excluded from differential analysis, OUTLIERS
 - Table of training-regulated features at 5% FDR, TRAINING_REGULATED_FEATURES
 - Bayesian graphical analysis inputs and results
 - Pathway enrichment of main graphical clusters, GRAPH_PW_ENRICH**
- Write code here: A pink box highlights the bottom right area of the editor, containing the text:

One important object in the data package is **PHENO**, which contains phenotypic information for every sample in the study. We can always get more information on any function or annotated dataset in R by using `?`, or typing PHENO in the help tab.

Note that `MoTrpacRatTraining6moData` must be installed and loaded with `library()` for this to work.

```
{r}
?PHENO
```**
- Results here: A blue box highlights the bottom left area of the editor, containing the console output:


```
> install.packages("~/Downloads/MoTrPAC-MotrpacRatTraining6moData-v2.1.0-0-gf831a4f.tar.gz", repos = NULL, type = "source")
Warning in untar2(tarfile, files, list, exdir, restore_times) :
  skipping pax global extended headers
* installing *source* package 'MotrpacRatTraining6moData' ...
** this is package 'MotrpacRatTraining6moData' version '2.0.0'
** using staged installation
** R
** data
*** moving datasets to lazyload DB
** inst
** byte-compile and prepare package for lazy loading
** help
*** installing help indices
** building package indices
```**
- Objects (data) here: A pink box highlights the top right area of the Environment pane, containing the text:

Values

| object_five | 5 |
|-------------|---|
| | |**
- Written files here: A pink box highlights the bottom right area of the Files pane, containing the text:

Files

| Name | Size | Modified |
|-----------------------------------|----------|--------------------|
| .. | | |
| R_Intro.Rmd | 44.9 KB | Jul 15, 2026, 9:13 |
| MoTrPAC-MotrpacRatTraining6moData | 380.2 MB | Jul 15, 2026, 8:33 |**

GitHub Commands via RStudio

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Git Pane

The screenshot shows the RStudio Git Pane interface. The top tab bar includes 'Environment', 'History', 'Connections', 'Git' (which is highlighted with a purple border), and 'Tutorial'. Below the tabs is a toolbar with icons for 'Diff', 'Commit', 'Pull', 'Push', 'History', and 'More'. On the right side of the toolbar, there are buttons for 'New Branch', a dropdown menu showing 'main', and a GitHub logo. The main area of the pane is divided into three sections: 'Staged', 'Status', and 'Path'. The 'Staged' section is currently selected and shows a list of files with checkboxes and status icons (yellow squares with question marks). The files listed are '.gitignore' and 'tutorial_GH_faculty.Rproj'.

| Staged | Status | Path |
|--------------------------|--------|---------------------------|
| ☐ | ?? | .gitignore |
| ☐ | ?? | tutorial_GH_faculty.Rproj |

GitHub Commands via RStudio

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Enables pulling from the original repo. This will sync the online repository with your cloned version.

Here is where you can push commits you have made within your clone to the online repo.

Will show you the history of your project (when files were committed, who was the last person to edit the repo, etc.).

This button allows you to create a separate branch from main. This enables more independent work.

Tells you which branch you are currently working in.

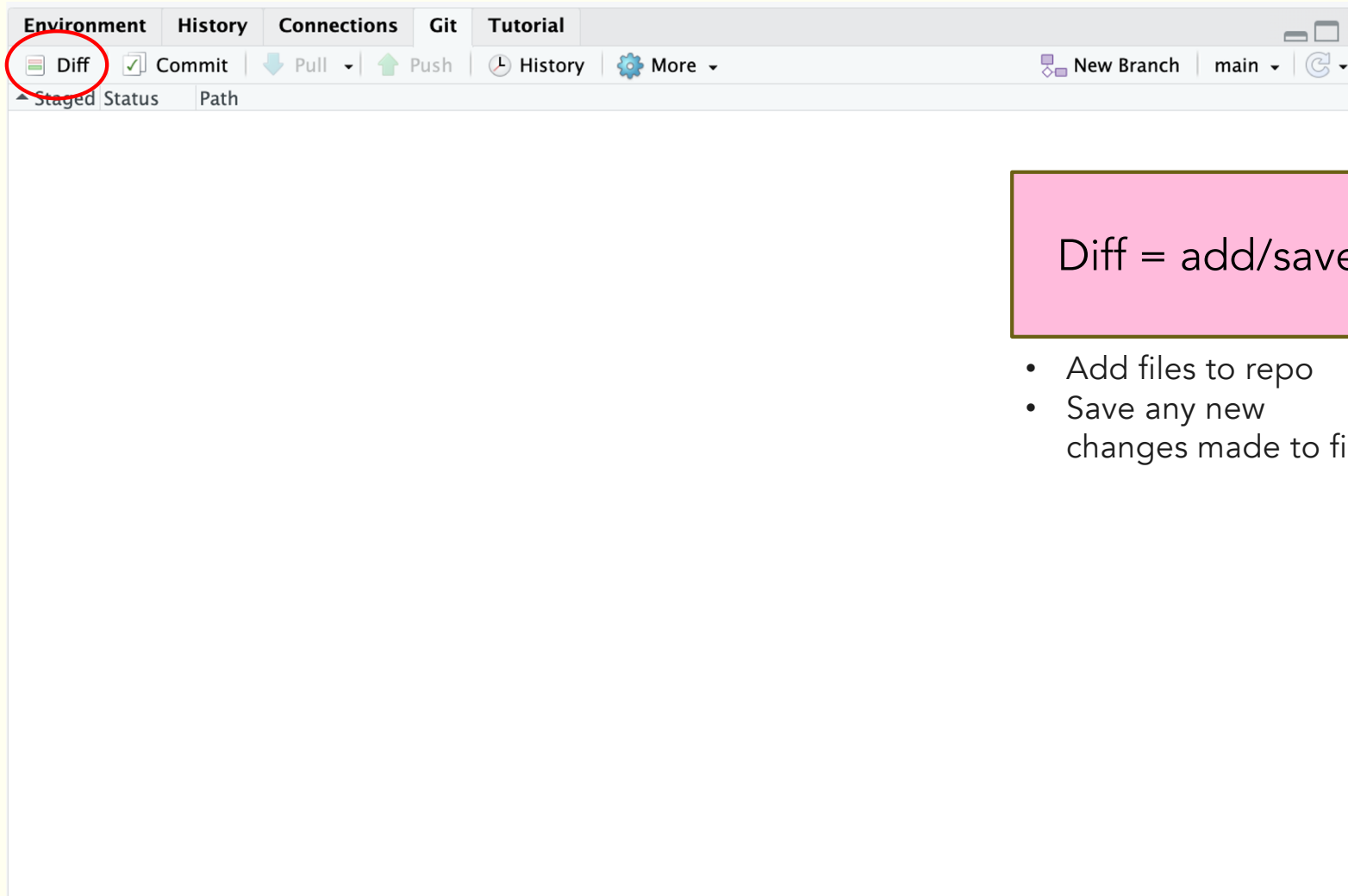
Allows you to commit changes you have made, preparing them to be pushed to the original repo.

Staged tells you if any changes have been committed. **Status** provides information on each file (has it been deleted, moved, renamed, etc).

As you add/delete/edit files, changes will appear in the panel!

The screenshot shows the RStudio interface with the following elements:

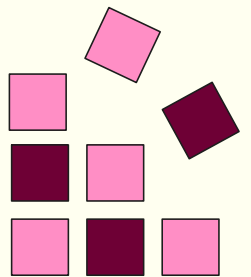
- Git Panel:** Contains buttons for Diff, Commit, Pull, Push, History, and More.
- Staged/Status Panel:** Shows a list of files with checkboxes and status icons (yellow question marks).
- Branch Selection:** A dropdown menu showing the current branch (main) and a button to create a new branch.



Diff = add/save

- Add files to repo
- Save any new changes made to files

20

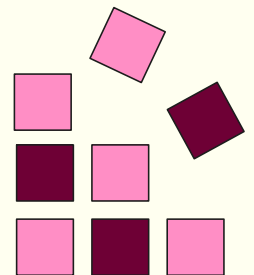


Diff Example

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Diff = add/save

- Add files to repo
- Save any new changes made to files



Say we wanted export our filtered dataset from our last workshop into a CSV:

Exporting Data

Let's export some filtered and manipulated data so it can be used in the future!

```
{r}
```

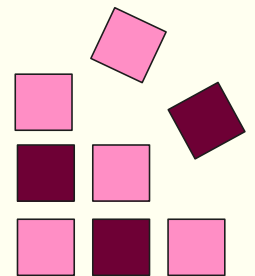
```
write.csv(pheno_dist, here("06_intro_to_R/worked_tutorial/distinct_pheno.csv"))
```

| Environment | | History | Connections | Git | Tutorial | | |
|--|--|--|---|---|---|---|------|
|  Diff |  Commit |  Pull |  |  |  |  | main |
| Staged | Status | Path | | | | | |
| ☐ | M | .DS_Store | | | | | |
| ☐ | ? ? | 06_intro_to_R/.DS_Store | | | | | |
| ☒ | A | 06_intro_to_R/worked_tutorial/R_Intro.Rmd | | | | | |
| ☒ | A | 06_intro_to_R/worked_tutorial/distinct_pheno.csv | | | | | |

We observe that the changed R notebook file and exported CSV shows up in the Git pane

Diff = add/save

- Add files to repo
- Save any new changes made to files



Changes | History | main | Stage | Revert | Ignore | Pull | Push

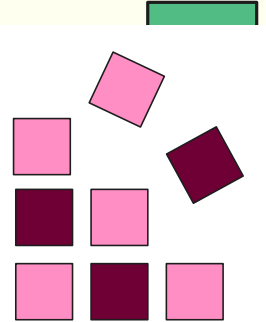
| Staged | Status | Path |
|-------------------------------------|--------|--|
| ☐ | M | .DS_Store |
| ☐ | ? ? | 06_intro_to_R/.DS_Store |
| ☐ | ? ? | 06_intro_to_R/worked_tutorial/R_Intro.Rmd |
| ☒ | A | 06_intro_to_R/worked_tutorial/distinct_pheno.csv |

Show | Staged | Unstaged | Context | 5 lines | Ignore Whitespace

```
838
839 ### Factor Re-leveling
840
841 The order of our groups doesn't quite make sense. It would be easier to read the graph if it were ordered from positive to negative rather than alphabetically. If we want to reorder
factors, we will use the `fct_relevel` function found within the [forcats](https://forcats.tidyverse.org/) package. We will take the `pheno_exp` object and use `mutate` to overwrite
the `group` variable with the `fct_relevel` function on the `group` variable. In this instance, we need to write the exact name and order of each factor by hand. We can check that
our reorganization worked by checking the `levels`.
842
843 ```{r}
844
845 pheno_exp <- pheno_exp %>% mutate(group=fct_relevel(group, "control", "1w", "2w", "4w", "8w" ))
846 levels(pheno_exp$group)
847 ```
848
849 Rerun the boxplots to see the controls show up first.
850
851 ```{r, solution=TRUE}
852 ggplot(pheno_exp, aes(x = group, y = Gak)) +
853   geom_boxplot() +
854   facet_wrap(~sex)
855 ```
856
857 # Exporting Data
858
859 Let's export some filtered and manipulated data so it can be used in the future!
860
861 ```{r}
862
863 write.csv(pheno_dist, here("06_intro_to_R/worked_tutorial/distinct_pheno.csv"))
864
865 ```
```

Diff = add/save

- Add files to repo
- Save any new changes made to files



Commit = describe the change

- Helps convey why the changes were made
- Confirm the changes you want to make

```

1  ---
2  titl
3  auth
4  outp
5  ht
6
7  ---
8
9  ## W
10
11 Hell
the
inte
(ima
the
the
or with the cursor in the grey code block, press control, shift, enter (⌘⇧⏎) simultaneously.
12
13 ```{r}
14 plot(cars)
15 ```
16
17 We see our results from that line of code immediately below it, which happens to be a graph about the speed and distance of different kinds of cars. Let's move on to the data we
will use for this workshop!
18
19 ## Introduction
20
21 This notebook introduces you to the `MotrpacRatTraining6moData` package, which contains the processed data for the Molecular Transducers of Physical Activity Consortium
(MoTrPAC) endurance training study in six-month-old rats. The goal of this notebook is to provide a beginner-friendly tour of the data.
22
23 ## Packages
24

```


Changes History main Stage Revert Ignore

| staged | Status | Path |
|-------------------------------------|--------|--|
| ☐ | M | .DS_Store |
| ☐ | ? | 06_intro_to_R/.DS_Store |
| ☒ | A | 06_intro_to_R/worked_tutorial/R_Intro.Rmd |
| ☒ | A | 06_intro_to_R/worked_tutorial/distinct_pheno.csv |

Pull Push

Commit message 46 characters

add completed tutorial and phenotype data csv.

☐ Amend previous commit

Commit

25

Commit = describe the change

- Helps convey why the changes were made
- Confirm the changes you want to make

Show Stage

```

1 ---
2 title
3 authc
4 outpu
5 htm
6 c
7 ---
8
9 ## We
10
11 Hello, and welcome to this workshop! We are using an R Notebook. An R Notebook is an R Markdown document with code blocks that can be executed independently and interactively, with the output visible immediately beneath the input. They are an implementation of [Literate Programming](https://en.wikipedia.org/wiki/Literate_programming) that allows for direct interaction with the R programming language while producing a reproducible document with publication-quality output. R notebooks can also be rendered using the knit button ![] (images/knit_button.png){width="53" height="23"} to create an HTML document that is easily viewable outside of R. We can see the difference between text and code as the text is in the white space and code blocks are in grey. To run any particular line of code, click the green play button ![] (images/play%20button-01.png){width="14" height="17"} in that chunk, or with the cursor in the grey code block, press control, shift, enter (⌘⇧⏎) simultaneously.
12
13 ```{r}
14 plot(cars)
15 ```
  
```

Unstage chunk

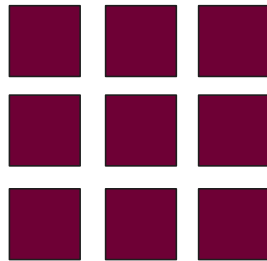
Unstage line

Commit message

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Commit =
describe the
change

- Helps convey why the changes were made
- Confirm the changes you want to make



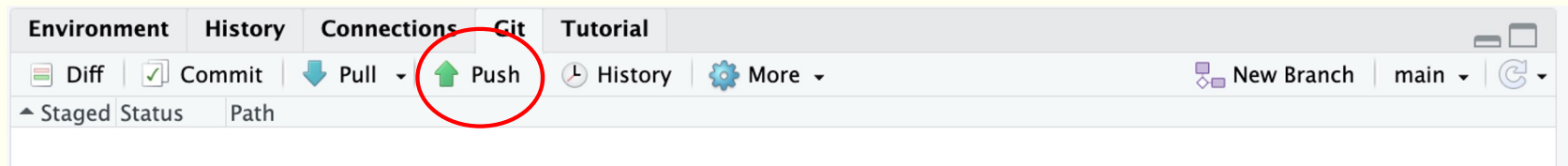
Git Commit

Close

```
>>> /usr/bin/git commit -F /var/folders/9s/mbqgj3bn1z1d6p_hdyj8lbd576jfx/T/RtmpFSj  
[main f71ccfe] add completed tutorial and phenotype data csv.  
2 files changed, 1013 insertions(+)  
create mode 100644 06_intro_to_R/worked_tutorial/distinct_pheno.csv
```

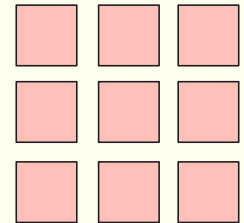
Storing files in the cloud

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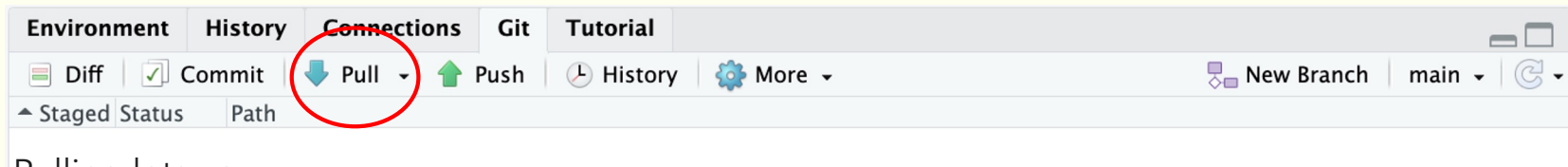
Push = send and store to the cloud

- Saves the new files on the cloud (GitHub)
- File changes are now documented



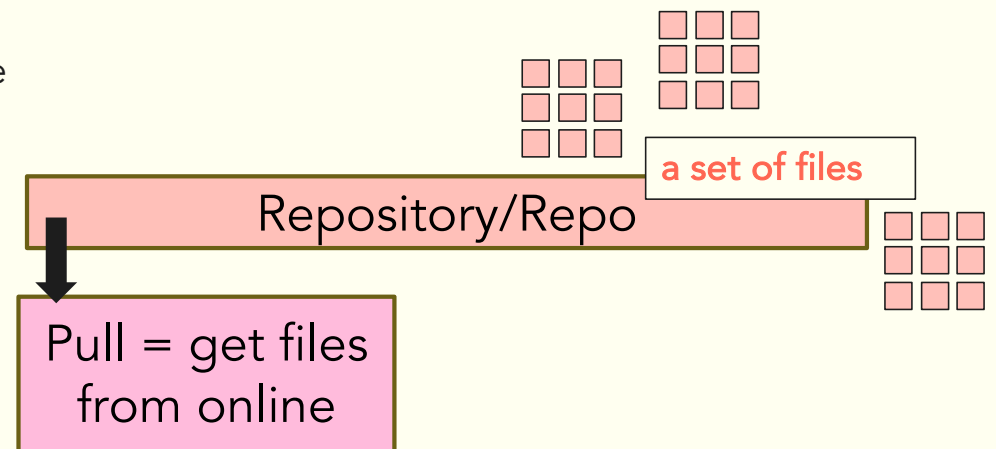
Refreshing your files from the cloud

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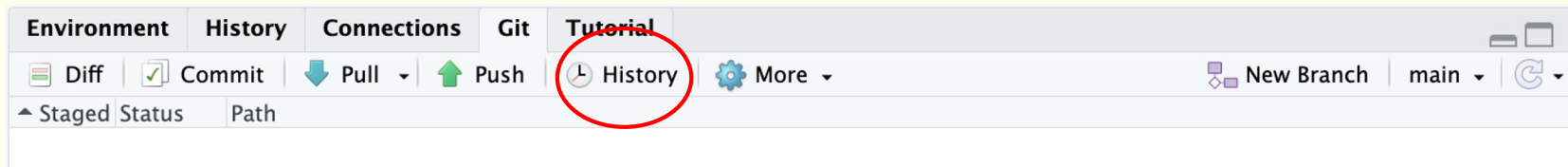
Pulling lets you:

- Access the latest updates to your files
- Pick up your project from any computer from where you left it
- Work with others and bring in any changes your teammates make



History in Git

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History shows you what changes
you've made to your project

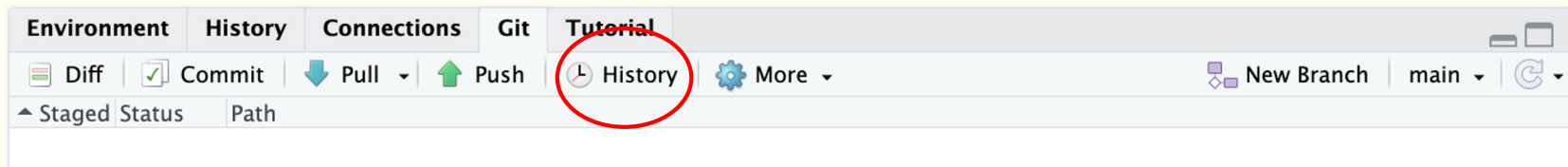
| Subject | |
|---|--|
|  | HEAD -> refs/heads/main origin/main origin/HEAD add completed tutorial and phenotype data csv. |

| | |
|------------|--|
| SHA | f71ccfe165880f67320760618fef3e07134013ec |
| Author | janedoe <janedoe@test.com> |
| Date (UTC) | 2026-07-15 18:35 |
| Subject | add completed tutorial and phenotype data csv. |
| Parent | 19b44a891326808683211d4dfc714a235ceca443 |

| | |
|---|--|
|  | 06_intro_to_R/worked_tutorial/R_Intro.Rmd |
|  | 06_intro_to_R/worked_tutorial/distinct_pheno.csv |

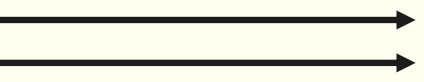
History in Git

30



History shows you what changes
you've made to your project

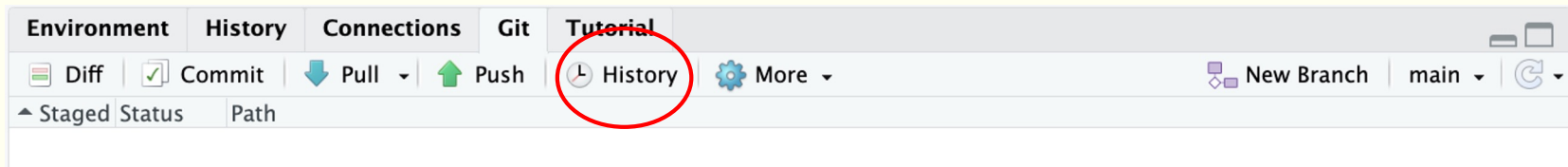
- You can also check who made the
change and when they were made



| Subject | |
|--|--|
| HEAD -> refs/heads/main origin/main origin/HEAD add completed tutorial and phenotype data csv. | |
| SHA | f71ccfe165880f67320760618fef3e07134013ec |
| Author | janedoe <janedoe@test.com> |
| Date (UTC) | 2026-07-15 18:35 |
| Subject | add completed tutorial and phenotype data csv. |
| Parent | 19b44a891326808683211d4dfc714a235ceca443 |
| |  06_intro_to_R/worked_tutorial/R_Intro.Rmd |
| |  06_intro_to_R/worked_tutorial/distinct_pheno.csv |

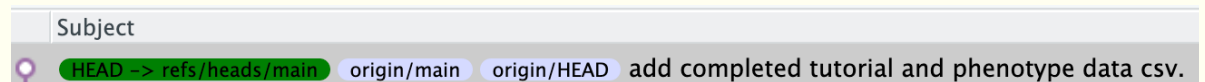
History in Git

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History shows you what changes you've made to your project

- You can also check who made the change and when they were made
- You can additionally look at what files were changed, added, or deleted



SHA f71ccfe165880f67320760618fef3e07134013ec
Author janedoe <janedoe@test.com>
Date (UTC) 2026-07-15 18:35
Subject add completed tutorial and phenotype data csv.
Parent 19b44a891326808683211d4dfc714a235ceca443

 06_intro_to_R/worked_tutorial/R_Intro.Rmd
 06_intro_to_R/worked_tutorial/distinct_pheno.csv

Getting Ready to Use GitHub

Step 0: Installations

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To complete this activity, you must have **Git**, **R**, and **RStudio** installed on your computing environment.

- [R/RStudio Download Instructions \(Windows & MacOS\)](#)
- [Git Install Instructions \(Various OS\)](#)

You must also have a **GitHub account**!
Github.com → Sign up

→ If you don't have the setup now, don't worry!

Register a GitHub account

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Simple Rules:

- **Use your real name** - like "sarah_johnson" or "mikec"
- **Keep it short** - easier to type and remember
- **Make it unique** - so no one else has it
- **Keep it professional** - something you'd be okay with teachers or future bosses seeing
- **Don't use temporary stuff** - avoid your school name or current city
- **Reuse from other places** - if you have LinkedIn or other professional social media accounts, use the same name

Remember: You'll use this username for years, so pick something you'll still like later!

Check Installations

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R

Open terminal and type:

```
R --version
```

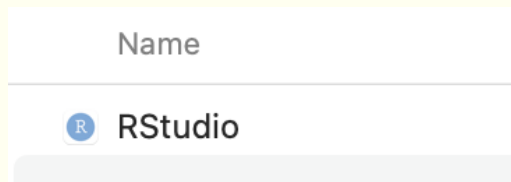
Example output:

```
(base) CP20034:~ janedoe$ R --version
R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical
Platform: aarch64-apple-darwin20
```

RStudio

Search for “RStudio” in your applications. You should see the RStudio app available for use.

Example output:



Git

Open terminal app and type:

```
git --version
```

Example output:

```
(base) CP20034:~ janedoe$ git --version
git version 2.39.5 (Apple Git-154)
```



Demo: Github Credentials, GitHub Repo Creation, & Clone Repo in RStudio



Step 1: Introduce yourself to Git

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Two ways:

Global configurations apply across repos at the user level.

1. Commands from the terminal

```
git config --global user.name "Jane_Doe"  
git config --global user.email "jane@example.com"  
git config --global --list
```

***substitute the name with your name and email associated with your GitHub account**

2. Using the *usethis* R package

```
install.packages(c("usethis", "gitcreds"))
```

```
Usethis::use_git_config(user.name = "Jane_Doe", user.email = "jane@example.org")
```

Step 2: Create a Personal Access Token (PAT)

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Two ways:

1. On Github:

- Click on your user icon in the upper righthand corner, and then navigate to **Settings/Developer Settings/Personal access tokens/Tokens (classic)**
- Click on Generate new token → Generate new token (classic)
- For scopes, select **repo** and **workflow**.
- Tips:

Set a time limit for your PAT!

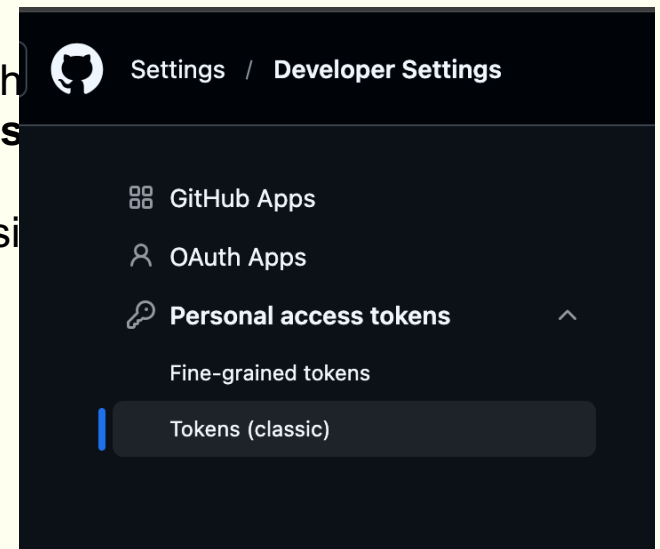
Keep it **SAFE / PRIVATE!**

Once you have generated your token, copy, paste, and **save it** immediately!

2. Using the *usethis* R package

```
usethis::create_github_token()  
gitcreds::gitcreds_set()
```

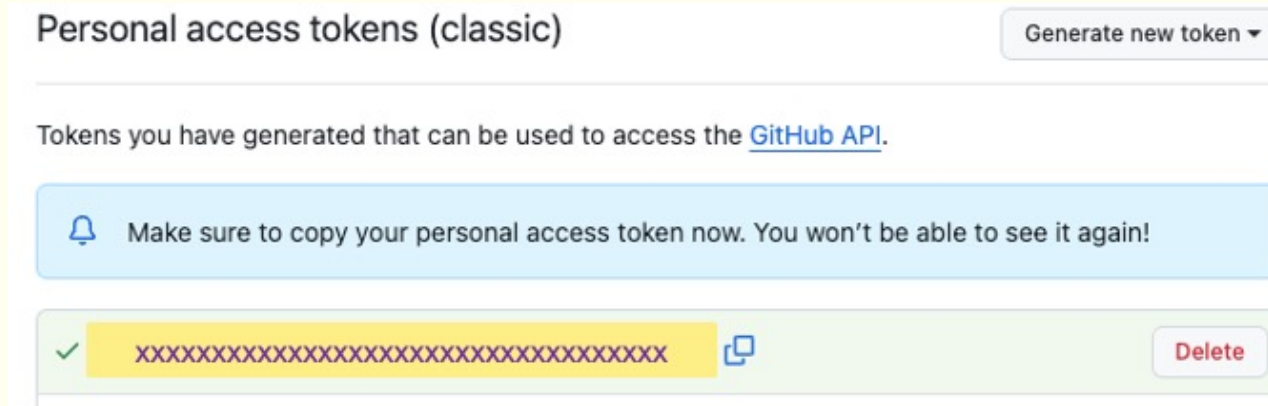
More details: usethis.r-lib.org/articles/git-credentials.html



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Using the *usethis* R package

```
gitcreds::gitcreds_set()
```



- Tips:

Set a time limit for your PAT!

Keep it SAFE / PRIVATE!

Once you have generated your token, copy, paste, and **save it** immediately!

More details: usethis.r-lib.org/articles/git-credentials.html

Step 3: Create Repo on Github

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Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * / Repository name *

 lakarstens /

Great repository names are short and memorable. How about [refactored-octo-umbrella?](#)

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

On ☒

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

Simple Rules:

- Make the name relevant to the task at hand
- We suggest something like:
`decode_institute_git_<your_name>`

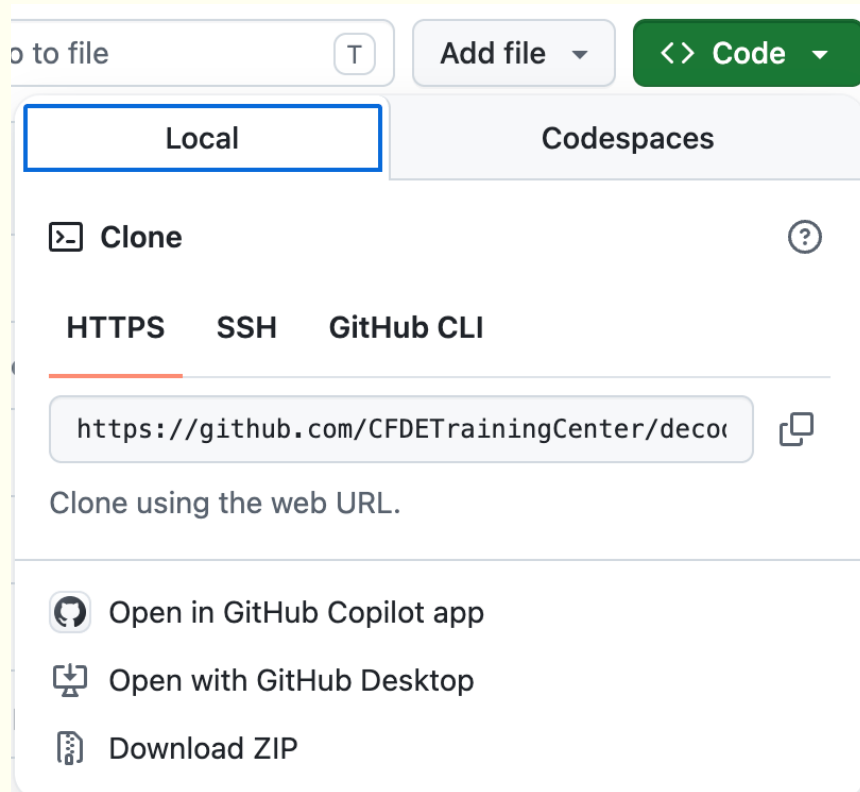
Remember: You'll want to name this something clear enough that you can find your work in the future.



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Step 4: Copy Repo URL

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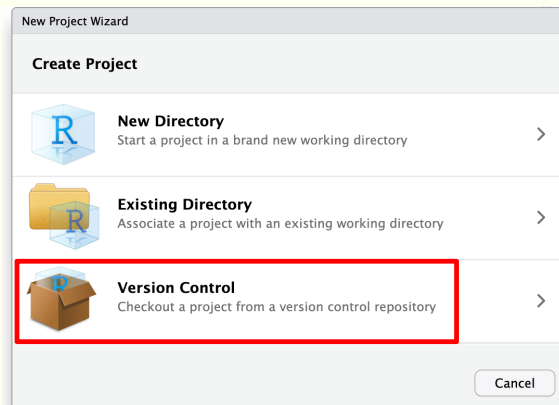


- Navigate to your Repo's home page
- Select Code
- Copy the HTTPS link to the repository.

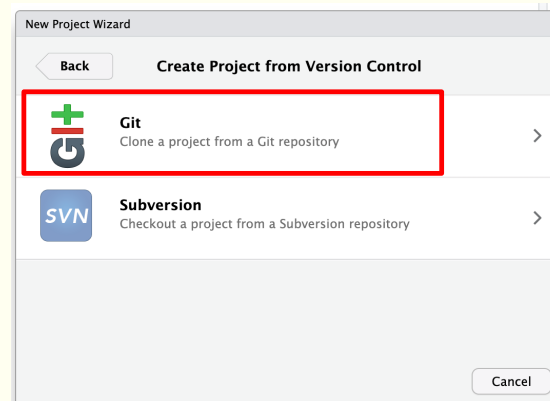
Step 5: Clone Repo

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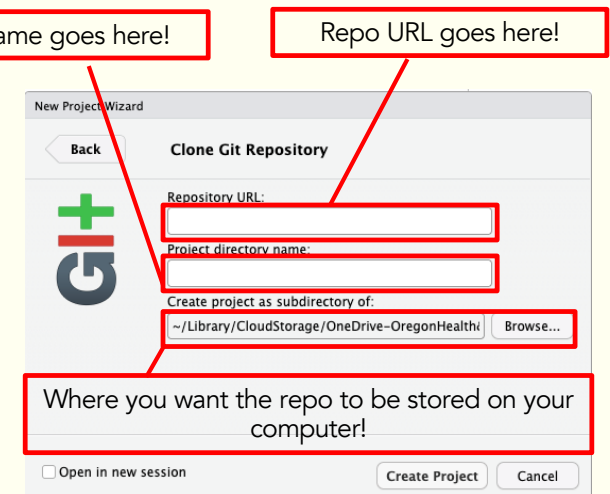
- Now that you have created a repo on GitHub, the next step is to clone (copy) that repo into RStudio.
- Begin in RStudio by clicking **File → New Project**.
- From there, follow the steps listed below:



Select Version Control



Click on Git



Fill in the information about the repository.

Note: If prompted for a password, **ENTER YOUR PAT,** instead of your account password.

Demo: Diff/Save > Commit > Push

Step 6: Create Code

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```
```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

## Example

```{r}

library(ggplot2)

data(mtcars)

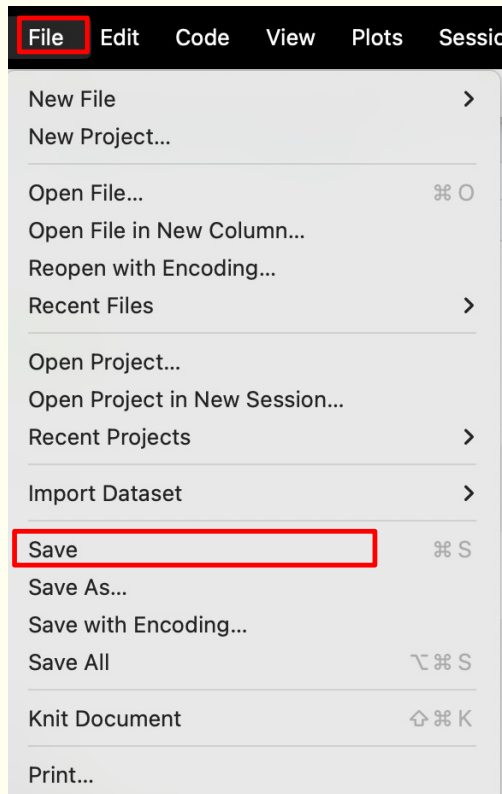
ggplot(mtcars, aes(x=factor(cyl))) +
 geom_bar() +
 xlab("Number of Cylinders") +
 ylab("Frequency")
```
```

- We are going to create a simple visualization using the mtcars dataset.
- This is a simple dataset that comes preloaded with R.
- The code on the left illustrates what the final product will look like.



Step 7: Save

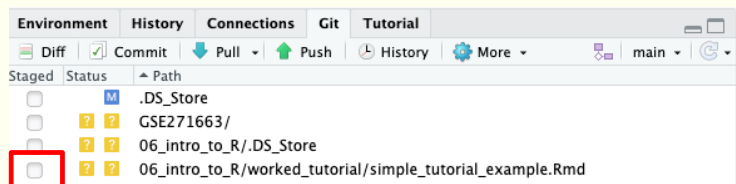
45



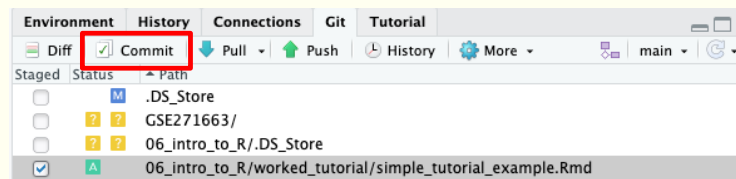
- Once you have completed the plot edits, it is time to save and rename your file.
- Click on File → Save
- Save the file
 <your_name>_cars_example.Rmd

Step 8: Stage + Commit

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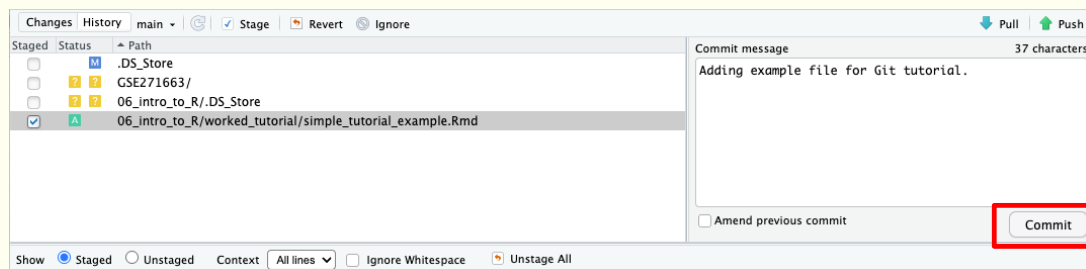


- To stage the file, click on the checkbox next to your file's name in the RStudio Git panel.
- Click on Commit
- Write a commit message explaining what changes you have made to your repo.
- When you're done, click on the Commit button



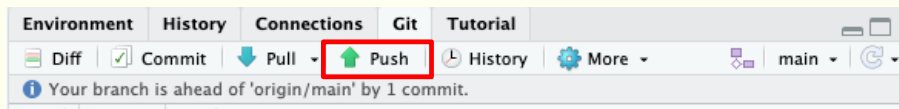
Commit message tips:

- Keep it short, to avoid cluttering the repo
- Summarize changes so they are clear to you when you look back at your work.



Step 9: Push

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- To complete the workflow, select the Push icon.
- This will sync the changes you made in RStudio to the original repo.
- An example of the type of message you should see in RStudio is shown on the left.
- To make sure everything worked, go back to your repo and see if your file is now visible!

Additional steps

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This is just the start!

Here's a few more common uses:

- Pull changes that were made by someone else

Ands lots of functionality we didn't cover:

- Add a collaborator
- Branching and pull requests
- Code review and feedback
- Issue tracking

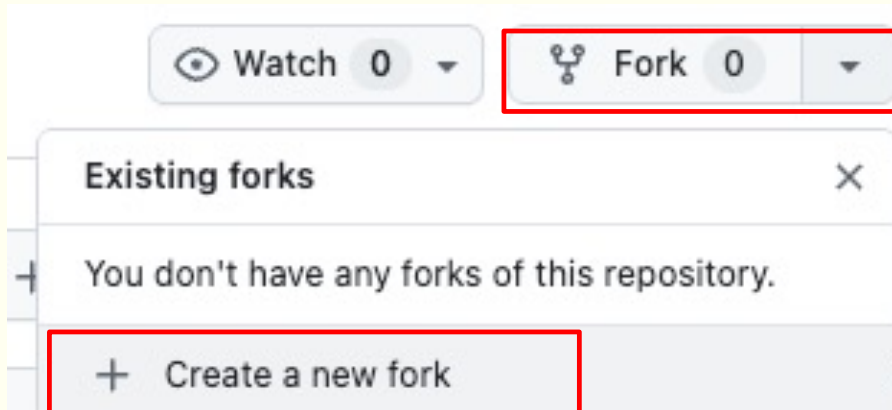


Your Turn!

Fork > Diff/Save > Commit > Push

Additional Activity: Fork a Repo

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- Navigate to the github.com/jdkudrick/decode-institute-2026-git-example
- Click on Fork → Create a new fork
- This will take you to a new page (similar to when you created your repository) where you will create your forked repo.

You can keep the name the same or change it – it's up to you!

*If you previously created a fork – **pull** to update it!*

Additional Activity: Modify a File

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Example Visualization

Below, we create a simple histogram using `ggplot`.

```
{r example_viz}
ggplot(pheno_select, aes(x = vo_max1)) + geom_histogram()
```

Your Turn!

Building off of the plot above, complete the following:

- Facet by sex of the mouse (call `facet_wrap()`)
- Fill in bars based on sex (`fill = sex`)

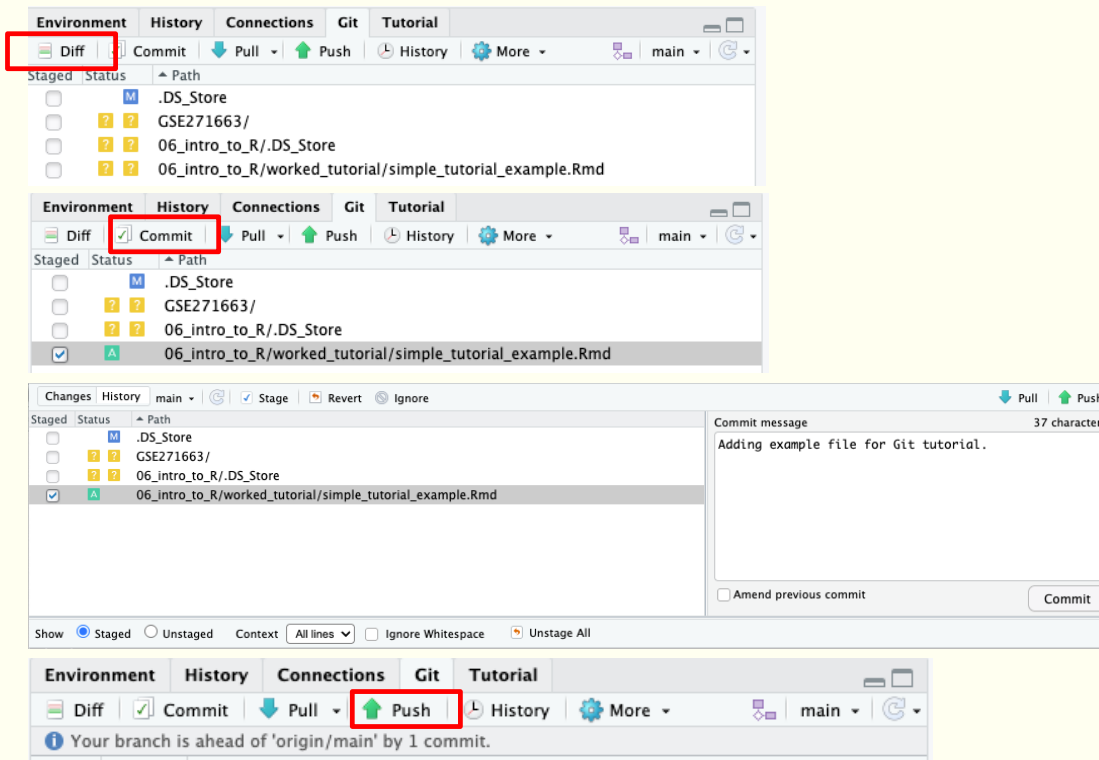
```
{r student_viz}
ggplot(pheno_select, aes(x = vo_max1)) +
  geom_histogram()
```

- Clone your forked repo into RStudio, following the same steps covered on previous slides.
- Open small_tutorial_example.Rmd
- The final code chunk (titled Your Turn!) instructs you to make edits to a plot.

This plot is visualizing some MoTrPAC data that was used during previous workshops

Additional Activity: Commit and Push

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- *Diff, Commit, and Push to your forked repo on github!*
- *Review on github once complete!*

Review of Key Terms

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| | |
|--------------------------|--|
| Repository (Repo) | All files and folders associated with a given project. These live online on GitHub. |
| Clone | Creating a copy of an online repository on your local device, allowing you to make edits locally. |
| Fork | Creating an online copy of a repository that you can edit without impacting the original repo. |
| Diff | Adding or modifying files to be tracked with git |
| Commit | Telling git to track the changes in a particular file |
| Push | Merging edits from your local device into the online repository. |
| Pull | Syncing changes from the online repository with your local version. |
| Issue | A way to report problems to the owner of a repository. Particularly helpful if using open-source packages. |
| Branch | Creating a separate channel where you can complete edits without modifying the main code (on the main branch). This allows for multiple people to work on the same projects at the same time, without overwriting one another. |



Github Desktop

Github Desktop

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GitHub Desktop

Experience Git without the struggle

Whether you're new to Git or a seasoned user, GitHub Desktop simplifies your development workflow.

- Gives you a simple interface to replicate the Git flow without commands

<https://github.com/apps/desktop>

Git

GitHub

Ex
st

Whether
simplifi

<https://github.com>

BioData

File Edit View Repository Branch Help

Current repository
desktop

Current branch
file-status-too... #17192 ✓

Pull origin
Last fetched 8 minutes ago 3 ↓

Changes 3

History

app\src\ui\lib\list\section-list.tsx

3 changed files

app\src\ui\lib\list\section-list.tsx

app\src\ui\lib\tooltip.tsx

app\src\ui\lib\tooltip-content.tsx

Stashed Changes

Bring `onRowKeyboardFocus` to `Se

Description

Co-Authors @sergiou87 @tidy-dev

Commit to file-status-tooltip

↑...

@@ -137,10 +137,19 @@ interface ISectionListProps {

137 137 source: IMouseClickSource

138 138) => void

139 139

140 + /** This function will be called when a row obtains focus, no matter how */

140 141 readonly onRowFocus?: (

141 142 indexPath: RowIndexPath,

142 143 source: React.FocusEvent<HTMLDivElement>

143 144) => void

145 +

146 + /** This function will be called only when a row obtains focus via keyboard */

147 + readonly onRowKeyboardFocus?: (

148 + indexPath: RowIndexPath,

149 + e: React.KeyboardEvent<any>

150 +) => void

151 +

152 + /** This function will be called when a row loses focus */

144 153 readonly onRowBlur?: (

145 154 indexPath: RowIndexPath,

146 155 source: React.FocusEvent<HTMLDivElement>

.....

↓

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Interface to
without

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Github Desktop

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- Gives you a simple interface to replicate the Git flow without commands

GitHub Desktop

Experience Git without the struggle

Whether you're new to Git or a seasoned user, GitHub Desktop simplifies your development workflow.

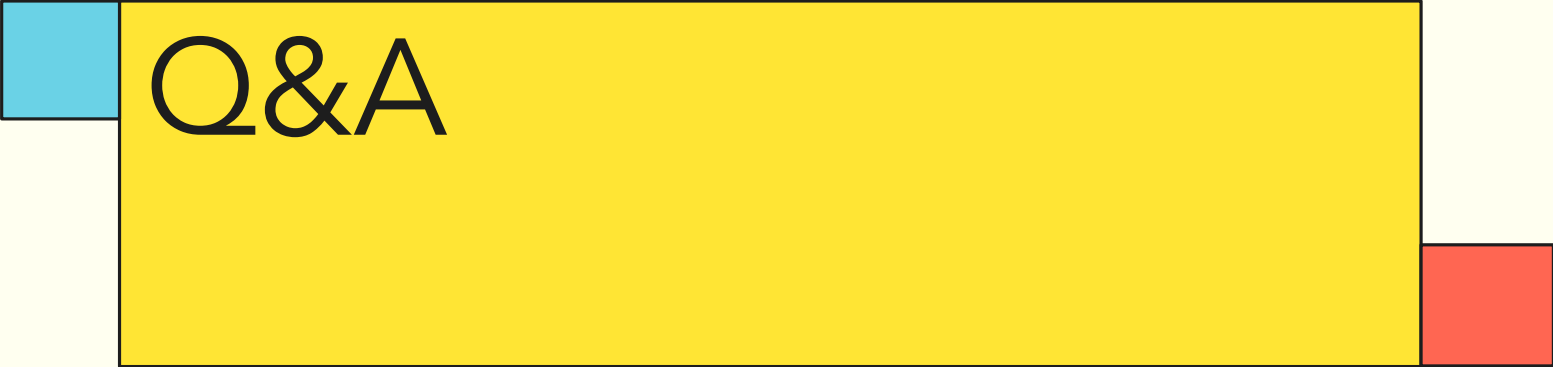
For tutorials and how to use GitHub Desktop:
<https://docs.github.com/en/desktop>

<https://github.com/apps/desktop>

Additional Resources

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1. Happy Git with R: <https://happygitwithr.com/>
2. GitHub's Git tutorial: <https://docs.github.com/en/get-started/git-basics/set-up-git>
3. Open and Reproducible Science in R: <https://oxford-ihtm.io/ihtm-handbook/>



Q&A



Thank you for
attending!